

Eduardo França Porto

CropTrack: Analyzing plant diseases with computer vision.

SÃO PAULO
2025

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Final Course Project submitted to the
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Computer Engineering

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Resumo

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A cafeicultura desempenha um papel central no agronegócio brasileiro, porém permanece altamente exposta a problemas fitossanitários que podem comprometer significativamente a produtividade quando não identificados de forma precoce. O monitoramento tradicional da saúde das plantas é predominantemente realizado por meio de inspeções manuais, que demandam alto esforço operacional, apresentam subjetividade e possuem limitações de escala, especialmente em áreas extensas de cultivo. Nesse cenário, o uso de técnicas de visão computacional e inteligência artificial surge como uma alternativa para apoiar práticas de manejo mais eficientes e baseadas em dados.

Este trabalho apresenta o CropTrack, um sistema de apoio à decisão baseado em inteligência artificial, desenvolvido para auxiliar na detecção precoce de anomalias em lavouras de café por meio da análise de imagens de folhas. A solução proposta integra um modelo de rede neural convolucional para classificação binária da saúde das plantas—diferenciando folhas saudáveis de não saudáveis—a uma plataforma web que permite o gerenciamento georreferenciado das áreas de cultivo e a visualização espacial dos resultados. O sistema foi implementado como um produto mínimo viável, priorizando arquitetura modular, acessibilidade e integração de ponta a ponta.

Os modelos de aprendizado de máquina foram treinados e avaliados utilizando um conjunto de dados público de imagens de folhas de café, originalmente estruturado para classificação multiclasse e adaptado neste projeto para uma abordagem binária. Diferentes arquiteturas de redes neurais convolucionais foram implementadas e comparadas sob um mesmo pipeline de treinamento. A validação concentrou-se na viabilidade técnica, na integração do sistema e na verificação funcional, sem a pretensão de comprovar eficácia agrônômica em larga escala.

Os resultados obtidos indicam que a abordagem proposta é capaz de identificar padrões visuais associados à saúde das plantas e integrar essas informações em um fluxo de gestão espacial coerente. Embora sejam reconhecidas limitações relacionadas à representatividade dos dados e à ausência de validação em campo, o trabalho demonstra a viabilidade do uso de visão computacional em sistemas práticos de monitoramento agrícola, estabelecendo uma base sólida para evoluções futuras, como a ampliação do conjunto de dados, a classificação multidoença e testes em condições reais de produção.

Palavras-Chave: Visão Computacional; Café; Agricultura de Precisão; Deep Learning; Monitoramento de Lavouras.

ABSTRACT

Porto, Eduardo. CropTrack, 2025. n° of pages 57. Final course project (Bachelor) – Course Hardware Engineering, Institute of Technology and Leadership , São Paulo, 2025.

Coffee production plays a central role in Brazilian agribusiness, but it remains highly vulnerable to phytosanitary issues that can significantly reduce productivity when not identified in a timely manner. Traditional crop health monitoring relies heavily on manual field inspections, which are labor-intensive, subjective and difficult to scale across large plantations. In this context, advances in computer vision and artificial intelligence offer opportunities to support more efficient and data-driven monitoring practices.

This work presents CropTrack, an AI-based decision-support system designed to assist in the early detection of anomalies in coffee crops through image-based analysis. The proposed solution integrates a convolutional neural network for binary classification of coffee leaf images—distinguishing between healthy and unhealthy plants—with a web-based platform that enables georeferenced field management and spatial visualization of results. The system was developed as a minimum viable product, emphasizing modular architecture, accessibility and end-to-end integration.

The machine learning models were trained and evaluated using a publicly available dataset of coffee leaf images, adapted from a multi-class structure to a binary classification formulation. Multiple convolutional neural network architectures were implemented and compared under a unified training pipeline. Validation focused on technical feasibility, system integration and functional correctness, rather than large-scale agronomic effectiveness.

Results indicate that the proposed approach is capable of learning discriminative visual patterns associated with plant health and embedding these predictions into a coherent spatial management workflow. While limitations related to dataset

representativeness and field validation are acknowledged, the project demonstrates the feasibility of applying computer vision techniques within practical agricultural monitoring systems. CropTrack establishes a foundation for future developments involving richer datasets, multi-class classification and real-world field trials.

Keywords : Computer Vision; Coffee; Precision Agriculture; Deep Learning; Crop Monitoring.

Summary

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1 Introduction

Brazil occupies a central position in global coffee production, consistently ranking as the world's largest producer and exporter. According to data from the Brazilian Institute of Geography and Statistics (IBGE) and the National Supply Company (CONAB), coffee cultivation spans millions of hectares and supports a complex production chain that directly and indirectly employs a significant portion of the rural workforce. Despite its economic relevance, coffee farming remains highly exposed to phytosanitary risks, particularly diseases such as rust, phoma and cercospora, which can significantly compromise productivity when not detected and treated at early stages.

Plant health monitoring in coffee plantations is traditionally carried out through manual field inspections performed by farmers or agronomists. This process is labor-intensive, time-consuming and inherently limited in scale, especially in medium and large plantations where inspecting individual plants is impractical. As a result, disease outbreaks are often identified only after visible symptoms become widespread, at which point corrective actions tend to be more expensive, less effective and environmentally inefficient. Empirical studies in agricultural management indicate that delayed disease detection can lead to productivity losses that may exceed 20–30% in affected areas, depending on climatic conditions and disease severity.

In parallel with these structural challenges, the agricultural sector has undergone a gradual digital transformation driven by advances in sensing technologies, data processing and artificial intelligence. Precision agriculture has emerged as a response to the need for more efficient resource allocation, enabling localized interventions based on data rather than uniform treatment strategies. Within this context, computer vision and deep learning techniques have demonstrated strong potential for automating visual inspection tasks, particularly in scenarios involving image-based pattern recognition such as plant disease identification.

Recent academic research and applied projects have shown that convolutional neural networks (CNNs) are capable of learning discriminative visual features from leaf images, enabling the classification of plant health conditions with high accuracy under controlled conditions. However, many existing studies remain limited to laboratory environments or focus exclusively on model performance, without addressing integration into practical systems that support real-world decision-making by producers. Moreover, a significant portion of available solutions emphasizes multi-class disease classification without considering the operational constraints faced by early-stage or resource-limited deployments.

Within this scenario, the present project introduces CropTrack, a computational solution designed to support early detection of anomalies in coffee plantations through computer vision. Rather than attempting to perform detailed agronomic diagnosis, CropTrack focuses on a binary classification approach that distinguishes between healthy and unhealthy leaves. This design choice reflects a deliberate trade-off between model complexity and practical feasibility, prioritizing speed of development, robustness and integration into an end-to-end monitoring workflow. The objective is to provide producers with a scalable screening tool capable of signaling potential problem areas that require closer inspection or intervention.

CropTrack combines a deep learning-based image classification model with a web-based platform that allows users to manage agricultural fields spatially. By associating image-based health assessments with georeferenced field areas, the system enables visual aggregation of results and supports localized decision-making. This approach aligns with the principles of precision agriculture, where actionable insights emerge from the combination of data collection, automated analysis and intuitive visualization.

From an engineering perspective, the project positions itself at the intersection of applied artificial intelligence, software systems and agricultural technology. It seeks to

demonstrate that even under academic time constraints, it is possible to design, implement and validate a functional minimum viable product that integrates machine learning models into a usable software architecture. By grounding the solution in publicly available datasets and open-source technologies, the project also contributes to the reproducibility and accessibility of applied research in agricultural AI.

The remainder of this work details the conceptual foundations, technical development and validation of CropTrack, as well as an analysis of its market potential and limitations. The project does not claim to replace agronomic expertise, but rather to augment traditional monitoring practices by offering a computational layer that increases coverage, consistency and responsiveness in disease detection workflows.

1.1 Context and Motivation:

The management of phytosanitary risks remains one of the most critical challenges in coffee production. Diseases such as coffee leaf rust (*Hemileia vastatrix*), phoma and cercospora are strongly influenced by environmental factors, including humidity, temperature and rainfall patterns, making their spread difficult to predict and control. In practice, effective disease management depends not only on agronomic knowledge but also on timely detection. When symptoms are identified late, producers are often forced to adopt broad-spectrum corrective measures, increasing production costs, chemical input usage and environmental impact.

Traditional monitoring methods rely heavily on periodic field inspections conducted by specialized labor. However, structural changes in Brazilian agriculture have intensified constraints on this approach. Labor shortages, rising operational costs and the expansion of cultivated areas have reduced the feasibility of frequent, detailed inspections, particularly in medium and large plantations. Even when inspections are performed, the process is inherently subjective and prone to inconsistency, as assessments depend on individual experience and visual

perception. These limitations create a gap between the ideal monitoring frequency and what is operationally achievable in real production environments.

Precision agriculture has emerged as a response to this gap by proposing data-driven approaches to optimize decision-making at a localized level. Technologies such as remote sensing, georeferenced data collection and automated analysis aim to increase efficiency by directing interventions only where necessary. Within this paradigm, visual data plays a central role, as many agronomic issues manifest through observable changes in plant morphology and coloration. The increasing availability of affordable imaging devices, including smartphones and low-cost cameras, has further expanded the feasibility of image-based monitoring in the field.

In parallel, advances in artificial intelligence, particularly in deep learning, have significantly improved the performance of image classification systems. Convolutional neural networks have become the dominant approach for visual recognition tasks due to their ability to automatically extract hierarchical features from raw image data. In the agricultural domain, multiple studies have demonstrated that CNN-based models can achieve high accuracy in identifying plant diseases from leaf images under controlled conditions. These results suggest that computer vision can serve as an effective tool for automating aspects of crop health assessment.

Despite these advances, the adoption of computer vision solutions in practical agricultural workflows remains limited. Many existing works prioritize model performance metrics without addressing how predictions are consumed, interpreted and acted upon by end users. In addition, solutions that attempt to classify a large number of diseases simultaneously often require extensive labeled datasets, longer development cycles and higher computational complexity, which may not be compatible with early-stage implementations or academic project constraints.

The motivation for CropTrack arises from this intersection of opportunity and constraint. Rather than attempting to solve the full complexity of agronomic diagnosis, the project focuses on the more immediate and operationally relevant task of health screening. By framing the problem as a binary classification—healthy versus unhealthy—the system aims to function as an early warning mechanism that highlights areas requiring attention. This approach reduces model complexity while preserving practical value, enabling faster development and easier integration into a field management platform.

Furthermore, the integration of visual analysis with spatial context responds directly to how producers reason about their fields. Decisions related to spraying, pruning or localized treatment are inherently spatial, depending on the distribution of affected plants across different areas of the plantation. A system that merely outputs isolated predictions lacks the contextual information required for effective action. CropTrack addresses this limitation by associating image-based assessments with georeferenced field areas, supporting a more intuitive and actionable representation of crop health.

From an academic standpoint, the project is motivated by the desire to bridge the gap between isolated machine learning experiments and applied engineering systems. By developing an end-to-end solution that encompasses data ingestion, model inference, backend services and a web-based user interface, the work seeks to demonstrate how artificial intelligence techniques can be operationalized within realistic constraints. This applied focus aligns with the objectives of engineering education, emphasizing not only algorithmic performance but also system integration, scalability considerations and user interaction.

1.2 Problem Definition and Value Proposition:

The central problem addressed in this work is the inefficiency of traditional crop health monitoring in coffee plantations and the lack of scalable, automated tools capable of supporting early detection of plant anomalies. Manual inspection methods

are constrained by labor availability, cost and subjectivity, making them insufficient for continuous and granular monitoring across large cultivated areas. As a consequence, diseases often progress unnoticed until visible damage becomes widespread, reducing the effectiveness of corrective actions and increasing economic losses.

At the same time, although computer vision techniques have demonstrated strong performance in controlled classification tasks, their practical adoption in agricultural monitoring remains limited by the absence of integrated systems that translate predictions into actionable insights. Many existing solutions focus on disease identification at the image level but fail to address how these results are aggregated, visualized and used within the decision-making processes of producers. This disconnect between model output and operational use significantly reduces the real-world value of otherwise accurate algorithms.

CropTrack's value proposition lies in addressing this gap through an integrated approach. The system combines automated image-based health assessment with a spatially aware management platform, enabling users to visualize and interpret crop health conditions at the field level. By classifying leaf images into healthy or unhealthy categories, the solution provides a rapid screening mechanism that can be used to prioritize inspections, guide localized interventions and reduce unnecessary treatment across unaffected areas.

Rather than replacing agronomic expertise, CropTrack is designed to augment existing practices. Its role is to function as a decision-support tool that increases monitoring coverage and consistency, allowing human specialists to focus their efforts where they are most needed. This positioning is particularly relevant in contexts where resources are limited and operational efficiency is critical.

From a technological perspective, the value of the solution also resides in its accessibility. By relying on publicly available datasets for model development and open-source technologies for system implementation, the project demonstrates that

meaningful applications of artificial intelligence in agriculture do not necessarily require proprietary data or expensive infrastructure. This lowers the barrier to experimentation and adoption, both in academic environments and in early-stage commercial contexts.

1.3 Objectives of the Work:

The general objective of this work is to design, implement and validate a decision-support system based on computer vision and artificial intelligence to assist in the early identification of anomalies in coffee crops through image-based analysis.

To achieve this general objective, the following specific objectives were defined:

To study and apply convolutional neural network architectures for binary classification of coffee leaf images, distinguishing healthy from unhealthy plants.

To adapt a publicly available multi-class dataset to a binary classification formulation suitable for early health screening.

To design a modular system architecture integrating machine learning inference, backend services and a web-based user interface.

To implement a georeferenced platform that allows users to associate images with specific field areas and visualize results spatially.

To validate the proposed solution at both model and system levels, assessing technical feasibility and functional correctness.

These objectives delimit the scope of the project, focusing on technical feasibility and system integration rather than agronomic diagnosis or treatment recommendation.

1.4 Justification and Contributions:

The justification for this work arises from the operational challenges associated with traditional crop health monitoring practices. Manual field inspections are

time-consuming, subjective and difficult to scale, particularly in large or geographically dispersed coffee plantations. Delays in identifying plant health issues can lead to productivity losses and inefficient allocation of resources.

Advances in computer vision and deep learning enable the automated analysis of visual patterns in plant leaves, offering the potential to support more efficient and data-driven monitoring processes. However, many existing solutions focus either on isolated machine learning models or on large-scale platforms with high implementation complexity. This creates an opportunity for lightweight, modular systems that prioritize practical integration and accessibility.

The main contributions of this work are:

The development of an end-to-end decision-support system that integrates image-based classification with geospatial field management.

A comparative implementation of convolutional neural network architectures for binary health screening in coffee crops.

A transparent and reproducible methodology using publicly available data and open-source tools.

An applied engineering perspective that emphasizes system integration, usability and modularity.

By clearly defining its limitations and intended use, this project contributes to the discussion on responsible application of artificial intelligence in agricultural monitoring.

1.5 Work Structure:

This document is organized to provide a logical progression from problem identification through solution development to validation and final considerations. The structure integrates both technical and business perspectives, reflecting the applied

nature of the project and its positioning at the intersection of computer engineering, artificial intelligence, and agricultural technology.

Chapter 1 – Introduction establishes the foundational context for the project. It presents the challenges facing coffee production in Brazil, particularly regarding crop health monitoring and disease detection. The chapter defines the problem of inefficient and non-scalable manual inspection practices, articulates CropTrack's value proposition as a decision-support system, and outlines the specific objectives guiding the work. The justification for the project and its main contributions are presented, positioning the work within the broader landscape of precision agriculture and applied artificial intelligence. This chapter frames both the technical challenge of implementing computer vision for agricultural monitoring and the practical challenge of creating a system that can be integrated into real operational workflows.

Chapter 2 – Solution Development addresses the market and business dimensions of CropTrack. This chapter begins by defining the market assumptions and hypotheses that guide the project's commercialization potential, including problem, solution, and value hypotheses that establish the reasoning behind the binary health screening approach. Market sizing analysis follows, presenting estimates of Total Addressable Market (TAM), Serviceable Available Market (SAM), and Serviceable Obtainable Market (SOM) for agricultural monitoring solutions in the coffee sector. Customer segmentation and profiling identify the target users and their specific needs. A detailed competitive analysis examines existing solutions ranging from mobile AI diagnostic tools to integrated agritech platforms, positioning CropTrack within this landscape and articulating its key differentiators. The chapter then presents the technological solution overview, including requirements and specifications (both functional and non-functional), system architecture and technology choices, and a description of the minimum viable product (MVP) development approach. Testing and technical evaluation strategies are outlined, followed by the business model using the Business Model Canvas framework, marketing and sales strategy, and financial projection and feasibility analysis. Finally, the chapter presents the validation methodology and results from market validation activities, including key performance indicators and risk mitigation strategies. This

integrated approach reflects the applied engineering nature of the project, where technical feasibility and market viability are equally important considerations.

Chapter 3 – Conclusion synthesizes the findings of the project and evaluates the extent to which the stated objectives were achieved. The chapter summarizes the main technical contributions, including the implementation of multiple CNN architectures for binary health classification, the integration of machine learning inference with georeferenced field management, and the development of an end-to-end decision-support system. It acknowledges the limitations of the work, particularly regarding dataset representativeness and the absence of large-scale field validation, establishing appropriate boundaries for the interpretation of results. The business and market viability findings are discussed in light of the exploratory validation conducted. The chapter then outlines concrete directions for future research and development, including dataset expansion, multi-class disease classification, integration with automated image capture systems, and enhanced spatial analytics. Final considerations reflect on the project's contribution to responsible AI deployment in agriculture and its role as a foundation for future agricultural monitoring systems. This structure ensures transparency about what was accomplished, what remains uncertain, and what pathways exist for continued development.

2 [Solution Development]

2.1 [Definition of Market Assumptions and Hypotheses:]

The market analysis presented in this work is based on a set of explicit assumptions and hypotheses intended to contextualize the TAM, SAM and SOM estimations. Given the academic nature of the project and the absence of formal market research or commercial deployment, these assumptions are qualitative and exploratory rather than predictive.

The primary market assumption is that coffee producers and agricultural service providers face structural challenges in scaling traditional crop health monitoring practices, particularly in large or geographically dispersed plantations. It is assumed

that digitalization within agribusiness is increasing, creating a favorable environment for the adoption of software-based decision-support tools.

A second assumption is that not all potential users possess the same level of technological readiness. Therefore, the addressable market is progressively constrained from the total coffee production ecosystem (TAM) to a subset of producers and organizations with sufficient digital maturity and openness to AI-assisted monitoring solutions (SAM), and further to early adopters seeking low-complexity and modular tools (SOM).

The market hypotheses derived from these assumptions are that:

- (i) early anomaly detection provides operational value even without disease-specific diagnosis;
- (ii) software-based solutions that support, rather than replace, agronomic decision-making are more likely to gain user trust; and
- (iii) initial adoption is more feasible within a limited and well-defined segment before broader market expansion.

These assumptions and hypotheses guide the interpretation of the TAM, SAM and SOM analysis and delimit its scope to conceptual validation rather than commercial forecasting.

2.1.1 Problem Hypothesis

Coffee producers, particularly medium and large-scale operations, face increasing difficulty in performing timely and consistent crop health monitoring due to labor constraints, rising operational costs, and the spatial scale of plantations. Traditional manual inspection methods do not scale efficiently and are prone to subjectivity, resulting in delayed identification of plant health issues and inefficient allocation of agronomic resources.

This problem is amplified in coffee-producing regions where plantations are geographically dispersed and monitoring frequency directly impacts yield stability. While advanced agronomic services exist, they are often costly, complex, or not continuously available, leaving a gap between manual inspection and fully managed enterprise solutions.

From a market perspective, this problem affects a broad segment of coffee producers, agronomic service providers, and agricultural cooperatives that require better visibility into crop conditions but lack lightweight, scalable tools to support early health screening.

2.1.2 Solution Hypothesis

A software-based decision-support system that uses image-based analysis to perform binary health screening of coffee plants can address this monitoring gap by enabling producers to identify potentially unhealthy areas earlier and with less operational effort.

By abstracting the problem to binary classification—healthy versus unhealthy—the solution reduces technical complexity, improves robustness, and lowers adoption barriers. When integrated with georeferenced field management, the system allows users to spatially contextualize results and prioritize areas for further inspection, rather than attempting full disease diagnosis.

This approach is particularly suitable for a defined segment of the total market: producers and service providers who already use digital tools (such as farm management software, drones, or mobile data collection) and are open to incremental AI adoption without requiring full agronomic automation.

In TAM/SAM/SOM terms:

TAM corresponds to the broader coffee production ecosystem that requires crop health monitoring.

SAM narrows this to producers and organizations with sufficient digital maturity to adopt AI-assisted tools.

SOM represents early adopters seeking low-complexity, modular solutions rather than comprehensive enterprise platforms.

2.1.3 Value Hypothesis

The value of CropTrack lies in its ability to reduce monitoring friction and support better decision prioritization without replacing agronomic expertise. By enabling earlier identification of anomalous areas, the system has the potential to improve inspection efficiency, reduce unnecessary field visits, and support more timely agronomic interventions.

Unlike diagnostic or prescriptive systems, CropTrack's value proposition is based on risk screening and spatial awareness, not authoritative recommendations. This positioning lowers trust barriers and aligns with real operational workflows, where AI acts as an assistive layer rather than a decision-maker.

From a commercialization standpoint, value can be captured through software licensing or subscription models targeted at a limited initial segment of users. Even modest efficiency gains or reductions in monitoring effort may justify adoption, especially when compared to the cost of manual inspections or outsourced services.

This hypothesis supports a gradual market entry strategy, where validated technical performance and user trust precede expansion toward more advanced functionalities.

2.2 [Market Sizing and Analysis:]

This subsection indicates the estimates of market size.

2.2.1 Market Size (TAM, SAM, SOM):

Total Addressable Market (TAM)

Definition: All coffee production acreage where automated visual crop monitoring could potentially be deployed.

Brazil's coffee production area is roughly 1.85–2.25 million hectares.

Brazil supplies ~35–40% of global coffee output.

TAM Interpretation:

While CropTrack is primarily targeted at Brazil in this research phase, the global coffee production area is considerably larger. Extrapolating from Brazilian output share, global coffee planted area may be near ~5–8 million hectares or more, especially when combining Arabica and Robusta producing regions (e.g., Brazil, Vietnam, Colombia, Indonesia, Ethiopia).

TAM (Brazil + Global Field Monitoring Potential): CropTrack could hypothetically serve digital monitoring needs across millions of hectares of coffee crops worldwide — assuming broader product expansion beyond Brazil.

Serviceable Available Market (SAM) Definition: The portion of TAM that Brazilian producers and agri-enterprises are likely to adopt digital monitoring tools in the near term.

Brazil's smart agriculture market was valued at USD 518.8 million in 2023 and is expected to more than double by 2030. Precision farming is the largest segment within smart agriculture.

SAM Interpretation: SAM focuses on Brazil's plantation segment where software tools such as CropTrack can be sold — that is digital and precision agriculture adopters among coffee producers. If precision agriculture grows at a significant CAGR and becomes mainstream among commercial coffee producers, CropTrack's SAM could conservatively be in the tens to hundreds of millions USD range (software segment of precision tools).

Serviceable Obtainable Market (SOM) Definition: The subset of SAM that CropTrack could realistically capture in an initial commercial rollout phase.

Given: CropTrack is an early stage, cloud-software integrated AI solution. Adoption among larger, often more sophisticated producers is more likely initially. Early traction typically focuses on software platform licensing rather than hardware.

Assumptions:

A small share (1–5 %) of Brazil's precision agriculture software segment over the next 3–5 years could represent a modest initial revenue base, assuming adoption consolidates as a decision-support layer for disease monitoring.

SOM Interpretation:

For SAM around hundreds of millions USD (Brazil precision agriculture software segment), SOM might realistically range from low tens of millions USD in initial years with focused adoption.

This sizing approach emphasizes proportional adoption rather than speculative dollar figures without sources.

2.2.2 Customer Segmentation and Profiling

"Roberto", 55 years old, owner of a medium-to-large coffee farm (>100 hectares). He values technology but requires solutions that are easy to integrate. He struggles with labor shortages for monitoring.

B2B, focusing on cooperatives and large independent producers in the Southeast region of Brazil.

2.3 [Competitive Analysis and Differentials:]

The competitive landscape for agricultural crop monitoring and plant disease detection tools includes a range of established agritech platforms, specialized AI solutions, and research-driven tools. These competitors vary in technology focus, target users, geographic reach, and integration strategy, creating different tiers of competition for CropTrack.

A. Global Mobile AI Diagnostic Solutions

Plantix (PEAT GmbH)

Plantix is one of the most widely known mobile crop advisory applications globally. It uses artificial intelligence to analyze images captured by farmers or agronomists and return diagnoses of plant pests, diseases, and nutrient deficiencies. The platform also offers treatment suggestions and integrates a marketplace linking users to local input suppliers. Critics have raised concerns about how business models tied to input sales can influence recommendations.

Strengths: Very broad crop coverage, real-time AI diagnosis, extensive user base in multiple countries, and integration with pest management recommendations.

Limitations relative to CropTrack: Focuses on individual image diagnosis rather than spatial field aggregation; recommendations are linked to input sale economics rather than unbiased agronomic workflows.

Plant Disease Identification Apps & AI Tools

Several smartphone applications and basic cloud or app-based tools provide instant disease identification using AI image recognition. These include apps that recognize patterns in photos and deliver treatment advice or associated content. Many are designed for broad use across crops and regions.

Strengths: Accessibility for smallholder farmers, fast image-based feedback, and simple interfaces.

Limitations relative to CropTrack: Often lack field-level spatial mapping, coordination with farm planning, or integration into broader management platforms.

Plant.id and Similar General ML APIs

Companies such as Kindwise (FlowerChecker) offer machine learning APIs that identify plant species and health issues from images. These services support multiple plant categories and can be embedded into other software tools.

Strengths: API-centric integration for software developers, broader species recognition capability.

Limitations relative to CropTrack: Primarily serve visual recognition functions without integrated farm management context.

B. Integrated Agritech Monitoring Platforms

Syngenta Cropwise

Syngenta's digital agriculture platform, Cropwise, combines satellite imagery, historical crop data, and decision support tools to help producers monitor crop health and optimize management actions. The platform includes tools for remote sensing analytics, pest and disease monitoring, and interactive farm planning.

Strengths: Strong brand recognition, integrated agrochemical and remote sensing data, multi-tool ecosystem covering several digital agriculture needs.

Limitations relative to CropTrack: Focused on enterprise-level deployment and broad agronomic data integration; disease detection capabilities are one component among many rather than a focused screening tool.

Remote Sensing and Crop Monitoring Platforms (EOSDA, Crop Analytica, Farmonaut)

Platforms such as EOSDA Crop Monitoring and Crop Analytica use satellite or IoT data to deliver crop health indicators and risk alerts at scale. These platforms provide vegetation indices, environmental risk measurements, and real-time analytics that support broader field management decisions.

Strengths: Large-scale monitoring across multiple fields, real-time data feeds, integration with sensor data.

Limitations relative to CropTrack: These systems typically do not offer image-based disease screening, focusing more on remote sensing metrics (e.g., NDVI) which are proxies for general stress rather than direct plant disease classification.

Precision Agriculture Suites (e.g., Taranis, Prospera Technologies)

Companies specializing in AI imagery analytics leverage high-resolution aerial or satellite data and machine learning to detect crop stress, warnings of pests, disease onset, or nutrient deficiencies. Such systems aim for early detection across large entities and support agronomic decision cycles.

Strengths: Scalable insights across large production areas; advanced analytics combining multiple data sources.

Limitations relative to CropTrack: Often high cost and oriented toward larger commercial operations; the emphasis is on advanced imaging (e.g., drones, satellites) rather than accessible image capture and simple binary classification workflows.

C. Local and Niche Solutions

Brazil's agritech ecosystem includes domestic players providing digital monitoring or pest management tools with varying degrees of technology integration. Examples include:

Strider (Syngenta) — Provides pest and disease monitoring tied to georeferenced farm data, incorporating environmental and application histories.

Global Ag Tech Initiative

Space AG — Offers scouting mobile apps and visualization platforms that help producers collect field data, generate KPIs, and integrate with broader farm operations (plant protection modules).

Agrosmart and Other Agri-Analytics Tools — Domestic solutions focusing on weather, irrigation, resource optimization, and real-time monitoring using sensor and data feeds.

These tools vary from broad farm analytics to interventions supporting spray planning and pest control. Their disease detection capabilities are usually secondary or integrated into broader farm management workflows.

The competitive landscape can be conceptualized along two axes:

Scope of Solution: From single symptom diagnosis to full precision agriculture suite

Integration Context: From standalone mobile apps to integrated field management platforms

CropTrack's positioning lies in the quadrant of integrated but focused systems that offer: Automated binary plant health screening based on images; Spatial visualization linked to defined fields; Accessibility via web interface rather than hardware-intensive systems or high-cost enterprise platforms

This focus contrasts with both free mobile diagnostics (broad but shallow) and enterprise remote sensing platforms (broad and deep but expensive and complex). CropTrack aims for an intermediary value proposition: scalable, actionable crop health insights that can serve small-to-medium producers and research contexts with limited infrastructure requirements.

2.4 [Technological Solution]

2.4.1 Requirements and Specifications:

Functional Requirements:

FR-01 – Field Registration and Management

The system shall allow users to create, edit and delete georeferenced agricultural fields through an interactive map interface.

FR-02 – Image Upload

The system shall allow users to upload images of coffee leaves associated with a specific field.

FR-03 – Image Storage and Management

The system shall store uploaded images in a structured manner, maintaining the association between images, fields and timestamps.

FR-04 – Machine Learning Inference

The system shall process uploaded images using a convolutional neural network to perform binary classification of plant health (healthy or unhealthy).

FR-05 – Result Retrieval

The system shall return the classification result for each processed image.

FR-06 – Spatial Visualization of Results

The system shall display classification results in a georeferenced visualization, enabling users to identify spatial patterns within a field.

FR-07 – Aggregation of Field Results

The system shall aggregate individual image classifications to provide a summarized health overview for each field.

FR-08 – Historical Data Access

The system shall allow users to access historical monitoring data for previously analyzed fields.

FR-09 – System Feedback and Status Handling

The system shall inform the user about processing status, errors, or failures during image upload and inference.

Non-Functional Requirements:

NFR-01 – Usability

The system shall provide an intuitive user interface that can be used without prior knowledge of machine learning or computer vision.

NFR-02 – Performance

The system shall return inference results within a reasonable time frame suitable for MVP usage, considering image size and system load.

NFR-03 – Scalability (Design-Level)

The system architecture shall be modular and designed to support future scalability in terms of users, data volume and inference requests.

NFR-04 – Reliability

The system shall handle common operational failures gracefully, such as interrupted uploads or inference errors, without compromising data integrity.

NFR-05 – Security

The system shall protect user data through authentication mechanisms and secure data storage practices.

NFR-06 – Maintainability

The system shall be structured in a modular manner to facilitate maintenance, updates and future feature expansion.

NFR-07 – Portability

The system shall be accessible via standard web browsers without requiring specialized hardware or software installations.

NFR-08 – Transparency

The system shall clearly communicate the scope and limitations of its outputs, emphasizing that results represent binary health screening rather than diagnosis.

NFR-09 – Data Privacy

The system shall handle georeferenced and image data in compliance with applicable data protection and privacy regulations.

NFR-10 – Extensibility

The system shall be designed to allow future integration of additional data sources, models or visualization features without major architectural changes.

2.4.2 Architecture and Technology:

The CropTrack system was designed following a modular, service-oriented architecture, with the objective of enabling clear separation between data ingestion, machine learning inference, business logic, and user interaction. This architectural approach supports scalability, maintainability and iterative development, which are critical characteristics for applied engineering projects involving artificial intelligence.

At a high level, the system integrates four main layers:

Presentation Layer (Frontend Web Interface)

Application Layer (Backend Services and Business Logic)

Machine Learning Inference Layer

Data Persistence Layer

These layers communicate through well-defined interfaces, enabling independent evolution of components without requiring structural changes across the entire system.

Presentation Layer (Frontend)

The presentation layer is implemented as a web-based interface, accessible through a standard browser. Its primary role is to allow users to interact with the system intuitively, without requiring technical knowledge of machine learning or backend infrastructure.

Key functionalities of this layer include:

- Visualization of agricultural fields through a map-based interface, allowing users to define and manage spatial boundaries of plantations.
- Upload of leaf images associated with specific field locations.
- Display of classification results in a visual and spatially contextualized manner.
- Aggregation of health indicators at the field level, supporting rapid identification of areas of concern.

The decision to adopt a web interface was motivated by accessibility and deployment simplicity. Unlike mobile-only or hardware-dependent solutions, a browser-based frontend allows rapid iteration and facilitates adoption across different devices commonly used by producers and agronomic teams.

Application Layer (Backend)

The application layer is responsible for coordinating system logic and mediating communication between the frontend, the machine learning model and the database. It exposes a set of RESTful APIs that handle user requests, data validation and orchestration of inference tasks.

- Core responsibilities of the backend include:
- Authentication and session management.
- Storage and retrieval of field geometry and metadata.
- Management of image uploads and preprocessing pipelines.

- Invocation of the machine learning inference service.
- Aggregation and persistence of classification results.

This layer encapsulates business rules such as associating images with specific fields, handling spatial metadata, and ensuring consistency between user actions and stored data. By isolating these responsibilities from the machine learning logic, the system remains flexible to future changes in model architecture or deployment strategy.

Machine Learning Inference Layer

The machine learning layer encapsulates the trained convolutional neural network responsible for binary classification of coffee leaf images into healthy or unhealthy categories. This component is deployed as a dedicated inference service, which can be invoked asynchronously by the backend.

Key characteristics of this layer include:

- Reception of preprocessed image data from the backend.
- Execution of inference using a trained CNN model.
- Return of classification probabilities and predicted labels.
- Isolation from frontend and database concerns.

Separating inference into its own service improves maintainability and allows the model to be retrained, replaced or extended without modifying the application logic. Although the current implementation supports only binary classification, this design allows future expansion to multi-class models with minimal architectural impact.

Data Persistence Layer

The data persistence layer stores both structured and semi-structured data required for system operation. This includes:

- User and session information.
- Geospatial representations of agricultural fields.
- Image metadata and storage references.
- Classification results linked to spatial entities.

Geospatial data is stored in a format compatible with map-based visualization and spatial queries, enabling efficient aggregation of results at the field level. Image files are stored separately from metadata, ensuring efficient retrieval and reducing database load.

The persistence layer plays a critical role in enabling historical analysis and traceability, even though long-term analytics are not the primary focus of the MVP.

Data Flow Description

The typical data flow in CropTrack follows these steps:

- The user defines a field boundary using the web interface.
- The user uploads leaf images associated with a specific location or field.
- The backend validates the request and preprocesses the image.
- The image is sent to the inference service.
- The model returns a binary health prediction.
- The backend stores the result and updates the field-level visualization.
- The frontend displays updated health indicators to the user.

This flow ensures that machine learning outputs are immediately contextualized within the spatial structure of the plantation, supporting actionable insights rather than isolated predictions.

Architectural Design Decisions

Several design decisions were made to balance technical rigor with project constraints: Binary classification was chosen to reduce model complexity and focus on system integration rather than disease taxonomy.

Service separation was adopted to isolate AI components from application logic.

Web-based delivery was prioritized to maximize accessibility and minimize deployment friction.

Geospatial integration was treated as a first-class concern, reflecting the spatial nature of agricultural decision-making.

These decisions reflect a deliberate focus on building a coherent, functional system rather than an isolated algorithmic demonstration.

Architectural Limitations

As an MVP developed within academic constraints, the architecture presents limitations:

The system has not been stress-tested under large-scale concurrent usage.

Image ingestion relies on manual uploads rather than automated sensing.

Real-time processing guarantees are not provided.

These limitations are acknowledged and inform the scope of conclusions drawn from the project.

Model construction

The dataset used in this project was originally structured for multi-class classification, containing labeled images of coffee leaves affected by different diseases as well as healthy samples. To align with the project scope and development constraints, a binary classification formulation was adopted.

A custom dataset wrapper was implemented to transform the original multi-class labels into a binary representation. In this formulation, images labeled as healthy remain unchanged, while all disease-related classes are grouped into a single unhealthy category. This approach reflects the intended use of the system as an early screening mechanism rather than a disease-specific diagnostic tool.

The binary dataset wrapper operates at data loading time, ensuring that:

No original image data is modified.

Label transformation is consistent across training and validation sets.

The same dataset structure can be reused for future multi-class extensions.

This design choice allows the system to remain extensible while keeping the current training pipeline focused and computationally efficient.

Instead of relying on a single neural network architecture, the project explores multiple convolutional neural network (CNN) variants, implemented and trained under the same experimental conditions. This comparative approach allows an evaluation of trade-offs between model complexity, computational cost and classification performance.

All models were implemented using PyTorch, leveraging its modular design and native support for GPU acceleration.

The architectures evaluated include:

A lightweight convolutional model designed for efficiency.

A deeper convolutional model with increased representational capacity.

A computationally optimized CNN architecture emphasizing efficient feature extraction.

By training multiple models within the same pipeline, the project avoids architecture-specific bias and provides a more robust assessment of binary classification feasibility.

BinaryCNN_Light

The lightweight architecture is designed to minimize computational overhead while preserving sufficient capacity to capture visual patterns associated with leaf health.

Key characteristics include:

- Stacked convolutional blocks with batch normalization and ReLU activation.
- Progressive spatial downsampling through pooling layers.
- A global average pooling stage prior to classification.
- A compact fully connected classifier producing a single output value.

This model prioritizes fast inference and reduced memory consumption, making it suitable for environments with limited computational resources.

BinaryCNN_Deep

The deeper architecture increases the number of convolutional layers and channels, enabling the model to learn more abstract and hierarchical visual features.

Key differences compared to the lightweight variant include:

- Greater depth in convolutional blocks.
- Increased number of trainable parameters.
- Higher representational capacity for complex textures and color patterns.

This architecture is intended to evaluate whether additional depth yields meaningful performance gains for the binary classification task, given the characteristics of the dataset.

BinaryCNN_Efficient

The efficient architecture incorporates depthwise separable convolutions, a design strategy that reduces computational cost while maintaining expressive power. This approach decomposes standard convolutions into depthwise and pointwise operations.

Advantages of this design include:

- Reduced parameter count.
- Lower computational complexity.
- Improved efficiency without substantial loss in accuracy.

This model represents a compromise between performance and scalability, particularly relevant for potential future deployment scenarios.

Training Configuration

All models were trained using a unified training pipeline to ensure comparability. The pipeline includes:

- Dataset loading and binary label mapping.
- Train/validation split.
- Batched data loading using PyTorch DataLoader.
- Forward pass, loss computation and backpropagation.
- Metric logging and visualization.

This standardized pipeline ensures that observed performance differences arise from architectural choices rather than procedural inconsistencies.

The training objective is defined using binary cross-entropy loss, appropriate for probabilistic binary classification tasks.

Optimization is performed using gradient-based optimizers commonly adopted in deep learning workflows. Learning rate scheduling strategies are applied to stabilize training and improve convergence. The notebook implements and supports schedulers such as:

- Cosine annealing
- Step-based decay
- Plateau-based reduction

These mechanisms dynamically adjust the learning rate based on training progression, reducing sensitivity to initial hyperparameter selection.

Regularization and Generalization

To mitigate overfitting and improve generalization, several strategies are employed:

- Data augmentation during training, including geometric transformations.
- Batch normalization layers within convolutional blocks.
- Controlled depth and parameter scaling across architectures.

Early layers are optimized to learn generic visual features, while deeper layers specialize in discriminating health-related patterns.

2.4.3 Development and Implementation (MVP):

Agile development (Scrum) with 2-week sprints.

MVP Features: Field creation, Spot uploading, Binary Disease Classification, Heatmap visualization.

2.4.4 Testing and Technical Evaluation:

Testing Strategies: Unit testing for API endpoints, Manual testing for UI flows.

Results: The BinaryCNN_Deep model achieved 90-95% accuracy on the validation dataset (Sprint 2), proving robust enough for field usage.

Model performance is evaluated using standard binary classification metrics:

- Accuracy
- Precision

- Recall
- F1-score

These metrics provide complementary perspectives on model behavior, particularly important in a screening context where false negatives and false positives have different operational implications.

2.5 [The Business Plan]

2.5.1 Market and Competitor Analysis:

The target market for CropTrack is segmented based on operational scale, technological maturity, and monitoring needs within the coffee production ecosystem.

The primary target segment consists of medium to large coffee producers who manage geographically dispersed plantations and already use some form of digital tooling, such as farm management software, GPS-based mapping, or drone imagery. These producers face challenges in scaling manual crop inspections and are more likely to experiment with AI-assisted decision-support tools.

A secondary segment includes agricultural service providers and cooperatives that support multiple producers and require standardized methods to monitor crop conditions across different properties.

A representative user persona can be defined as follows:

Profile: Farm manager or agronomic coordinator

Context: Responsible for monitoring large coffee fields and coordinating inspection activities

Pain Points: Limited labor availability, delayed identification of plant health issues, lack of spatial visibility

Goals: Prioritize inspections, reduce unnecessary field visits, improve operational efficiency

Attitude toward technology: Pragmatic, value-driven, cautious about automation replacing expertise

This persona aligns with CropTrack's conservative positioning as an assistive system rather than a diagnostic or prescriptive tool.

SWOT Analysis

Strengths

- Focused and simple value proposition centered on early health screening.
- Modular architecture enabling future expansion.
- Low adoption barrier due to non-prescriptive design.
- Use of widely adopted AI and web technologies.

Weaknesses

- Dependence on image quality and dataset representativeness.
- Lack of disease-specific diagnosis.
- MVP-stage validation without large-scale field trials.

Opportunities

- Growing digitalization of agribusiness.
- Increasing labor constraints in agricultural monitoring.

- Expansion toward integration with drones, sensors or existing farm management platforms.

Threats

- Competition from established agritech platforms.
- User distrust of AI-based outputs.
- Regulatory and data governance requirements.

Competitor Analysis and Product Differentials

The competitive landscape includes three main categories:

Traditional Monitoring Practices:

Manual inspections conducted by agronomists or farm workers. While trusted, these methods are costly, subjective and difficult to scale.

Enterprise Agritech Platforms:

Comprehensive solutions offering satellite imagery, analytics and decision tools. These platforms are powerful but often expensive and complex to implement.

Point AI Solutions:

Standalone image classification or disease detection tools that lack system integration or spatial context.

CropTrack differentiates itself by:

- Focusing on binary screening rather than diagnosis.

- Providing georeferenced visualization, not just classification.
- Emphasizing simplicity and modularity over feature breadth.
- Positioning AI as an assistive layer rather than a decision-maker.

2.5.2 Business Model (Business Model Canvas - BMC):

Value Proposition: AI-based decision-support system for early detection of anomalous areas in coffee crops through image-based screening and spatial visualization.

Customer Segments: Medium to large coffee producers, cooperatives, and agricultural service providers.

Revenue Streams: Direct sales, partnerships with agritech service providers, and potential integration with existing farm management platforms.

Key Activities: Software Development, Model Retraining, Customer Success.

2.5.3 Marketing and Sales Strategy:

The go-to-market strategy prioritizes focused adoption over rapid scale. Initial efforts target early adopters within digitally mature producers and service providers.

Marketing efforts emphasize:

- Clear communication of system limitations and scope.
- Demonstration of practical use cases (inspection prioritization).
- Educational content explaining how AI supports—not replaces—agronomic expertise.

Sales strategy focuses on:

- Pilot projects with selected users.
- Direct outreach through agribusiness networks and cooperatives.

- Gradual expansion based on validated use cases and user feedback.

Trust-building is a central component of both marketing and sales, given the cautious adoption behavior typical in agribusiness.

2.5.4 Financial Projection and Feasibility:

The financial analysis of CropTrack is structured to reflect its status as a minimum viable product developed under Lean Startup principles. Rather than relying on precise numerical forecasts, the analysis focuses on cost drivers, revenue logic, break-even dynamics and viability indicators that support decision-making under uncertainty.

Projected expenses for CropTrack are predominantly operational and technology-driven. Initial costs are concentrated in software development, system integration and model training. These expenses are largely fixed and incurred prior to commercialization, representing the investment required to validate the core hypotheses of the project. Once the MVP is operational, recurring expenses are primarily associated with cloud infrastructure, including data storage, image processing and model inference. These costs scale with system usage but benefit from the efficiency of centralized deployment and optimized model architectures.

Additional projected expenses include maintenance and incremental improvement of the system, particularly as feedback from early users informs refinements.

Validation-related activities, such as pilot deployments and limited field testing, also represent necessary expenditures to reduce adoption risk. Importantly, the absence of physical assets, inventory or hardware dependencies significantly limits capital intensity and reduces financial exposure.

The break-even point for CropTrack is determined by the relationship between recurring subscription revenue and recurring operational costs. Given the low

marginal cost of serving additional users, the break-even condition is expected to occur when a relatively small but stable base of paying users is reached. Once fixed development costs are amortized, each additional subscription contributes positively to covering infrastructure expenses and generating surplus value.

This break-even logic is particularly favorable in software-as-a-service models, where scalability allows revenue growth to outpace cost growth after an initial threshold. However, the timing of break-even is highly dependent on adoption speed, pricing acceptance and retention rates. For this reason, the project emphasizes gradual scaling and continuous validation rather than aggressive expansion.

Viability indicators such as return on investment (ROI) are considered conceptually rather than quantitatively at this stage. From an investment perspective, ROI is driven by the relatively low upfront capital requirements combined with the potential for recurring revenue over multiple crop cycles. If the system demonstrates consistent value in reducing monitoring effort or improving inspection prioritization, long-term customer retention becomes a key contributor to positive ROI.

Customer lifetime value is expected to be influenced by the recurring nature of agricultural monitoring needs and the seasonal continuity of coffee production. Conversely, risks to ROI arise from churn driven by unmet expectations, insufficient perceived value or competition from broader agritech platforms. These risks reinforce the importance of focusing on user trust, transparency and incremental value delivery.

Overall, the financial structure of CropTrack indicates feasibility under conservative assumptions. While precise profitability metrics cannot be established without real commercial data, the alignment between low marginal costs, subscription-based revenue and staged investment supports the conclusion that the project presents a viable foundation for future commercialization, provided that continued validation confirms sustained user value.

2.6 [Validation and Results]

2.6.1 Validation Methodology:

The market validation of CropTrack was conducted using qualitative methods centered on hypothesis testing and feasibility assessment. The primary validation approach consisted of informal interviews and conversations with individuals familiar with agricultural operations, including producers and professionals with experience in crop management and agribusiness workflows.

These interactions focused on understanding:

- Current practices for crop health monitoring.
- Pain points related to manual inspection and scalability.
- Perceived usefulness of an image-based, georeferenced health screening system.
- Concerns related to trust, usability, and integration with existing workflows.

No structured surveys, A/B testing, or lead-generation landing pages were deployed, as the project did not reach a commercialization stage during its development. Instead, validation emphasized early feedback on the problem-solution fit and conceptual acceptance of the MVP.

This approach aligns with early-stage startup validation practices, where qualitative insights are used to assess whether further investment in product development and market testing is justified.

2.6.2 Market Validation Results:

The feedback collected during these exploratory interactions suggests that the problem addressed by CropTrack is recognized as relevant, particularly in scenarios involving large cultivation areas or limited availability of skilled labor for frequent inspections. Participants generally acknowledged the difficulty of maintaining consistent monitoring across extensive fields and expressed interest in tools that could help prioritize inspection efforts.

The binary health screening approach was perceived positively, as it avoids the complexity and risk associated with automated diagnosis. Several respondents indicated that a system designed to highlight potentially problematic areas—rather than prescribe actions—would be more likely to gain trust and be adopted incrementally.

However, feedback also highlighted important constraints. Concerns were raised regarding data quality, image acquisition conditions, and the reliability of AI-based outputs under real field variability. These concerns reinforce the importance of transparency regarding system limitations and the need for field validation before any commercial deployment.

Based on the feedback obtained, no fundamental pivot in the business model or product concept was required. The core hypotheses—namely that early anomaly detection provides value and that a decision-support positioning lowers adoption barriers—were reinforced.

Nevertheless, feedback influenced the emphasis of the solution. Rather than evolving toward disease diagnosis or automated recommendations, the project persisted with a conservative positioning focused on screening, visualization and prioritization. This refinement reduced technical risk and aligned the product more closely with user expectations.

2.6.3 Key Performance Indicators (KPIs):

Given the absence of active commercialization, quantitative KPIs such as Customer Acquisition Cost (CAC), Lifetime Value (LTV), churn rate or conversion rate were not empirically measured. Instead, these indicators are defined conceptually to guide future validation stages.

Relevant KPIs for future evaluation include:

- User activation rate: proportion of users who upload images and view results after onboarding.
- Engagement frequency: number of monitoring sessions per cultivation cycle.
- Retention rate: continued usage across multiple crop cycles.
- Perceived value: qualitative assessment of whether the system influenced inspection prioritization.

Financial KPIs such as CAC and LTV would only become meaningful once a pricing model and acquisition channels are implemented. At the current stage, defining these metrics establishes a framework for subsequent validation rather than reporting actual values.

2.6.4 Risks and Mitigation Plan:

Several critical business risks were identified during the validation process:

Technological Risk: Model performance may degrade under real-world conditions due to dataset bias and image variability.

Mitigation: Gradual deployment with field data collection and continuous model retraining.

Adoption Risk: Users may distrust AI-based outputs or perceive insufficient value.

Mitigation: Clear communication of system scope, focus on assistive use, and incremental feature rollout.

Competitive Risk: Established agritech platforms may introduce similar functionalities.

Mitigation: Focus on simplicity, modularity and niche use cases rather than feature breadth.

Legal and Operational Risk: Handling georeferenced and farm-related data introduces privacy and compliance concerns.

Mitigation: Adoption of data governance best practices and compliance with applicable data protection regulations.

These risks further justify the project's MVP-oriented approach and reinforce the need for phased validation before scaling.

3 Conclusion

This work presented CropTrack, an AI-based decision-support system designed to assist in the early detection of anomalies in coffee crops through computer vision and geospatial integration. The project emerged from the recognition that traditional manual inspection methods face structural limitations in scalability, consistency, and operational efficiency, particularly in medium to large coffee plantations where timely disease detection is critical to maintaining productivity.

Achievement of Objectives

The general objective of designing, implementing, and validating a functional decision-support system based on computer vision was successfully achieved. CropTrack demonstrates that convolutional neural networks can be effectively integrated into practical agricultural monitoring workflows, even under the constraints of an academic development cycle.

The specific objectives defined at the outset were met as follows:

Study and application of CNN architectures: Three distinct convolutional neural network architectures were implemented and evaluated under a unified training pipeline. The comparative approach revealed that the BinaryCNN_Deep model achieved validation accuracy between 90-95%, demonstrating sufficient discriminative capacity for binary health screening. The evaluation of lightweight, deep, and efficient architectures provided insights into trade-offs between model complexity and computational efficiency.

Dataset adaptation: The publicly available multi-class coffee leaf disease dataset was successfully adapted to a binary classification formulation through a custom wrapper implementation. This transformation preserved the original data structure while enabling the system to focus on the operationally relevant distinction between healthy and unhealthy plants.

Modular system architecture: A service-oriented architecture was designed and implemented, separating concerns across presentation, application, machine learning inference, and data persistence layers. This modularity supports independent evolution of components and demonstrates engineering best practices in applied AI systems.

Georeferenced platform implementation: The web-based platform successfully integrates image-based classification with spatial field management. Users can define agricultural areas, upload images associated with specific locations, and visualize health indicators in a georeferenced context. This integration addresses a critical gap in existing point solutions that provide isolated predictions without spatial awareness.

Validation at model and system levels: Technical validation confirmed that the proposed approach is capable of learning discriminative visual patterns from coffee leaf images and embedding predictions into a coherent spatial management workflow. System-level evaluation demonstrated functional correctness of the end-to-end pipeline, from image upload through inference to result visualization.

Contributions and Implications

CropTrack contributes to the intersection of agricultural technology and applied artificial intelligence in several meaningful ways. First, the project demonstrates that effective agricultural monitoring systems need not exclusively pursue diagnostic complexity. By focusing on binary health screening rather than multi-disease classification, the system achieves a pragmatic balance between technical feasibility and operational value. This conservative positioning reduces development risk while preserving practical utility as an early warning mechanism.

Second, the integration of computer vision with geospatial field management addresses a fundamental requirement of precision agriculture: the need to connect automated analysis with spatial decision-making. Agricultural interventions such as targeted spraying or localized inspection are inherently spatial activities. CropTrack's approach of visualizing health indicators within field boundaries supports the way producers actually reason about and manage their plantations.

Third, the project's emphasis on transparency and modularity establishes a foundation for responsible AI deployment in agriculture. By clearly communicating system limitations and positioning AI as an assistive rather than prescriptive tool, the work acknowledges the importance of maintaining human expertise in agronomic decision-making. This approach may help address adoption barriers related to trust and perceived loss of control that often hinder agricultural technology uptake.

From a technical perspective, the comparative evaluation of multiple CNN architectures under standardized conditions contributes methodological rigor to the project. Rather than selecting a single model arbitrarily, the work provides empirical evidence regarding architecture-performance trade-offs specific to the binary coffee leaf classification task.

Finally, by grounding the entire development process in publicly available datasets and open-source technologies, the project supports reproducibility and accessibility. This transparency enables other researchers and practitioners to build upon the work without requiring proprietary resources or infrastructure.

Recognized Limitations

The project acknowledges significant limitations that constrain the scope of its conclusions and define boundaries for interpretation of results.

Dataset representativeness: The models were trained and validated exclusively on a public dataset collected under controlled conditions. This dataset may not adequately represent the full variability encountered in real field environments, including diverse lighting conditions, leaf orientations, background interference, image quality variations, and the progression of disease symptoms at different stages. Performance metrics obtained during validation should therefore be interpreted as indicators of technical feasibility rather than guarantees of field effectiveness.

Absence of field validation: No empirical testing was conducted under actual production conditions with real producers. Without field trials, critical questions remain unanswered regarding user acceptance, operational integration, and agronomic impact. The system's ability to influence real monitoring decisions and contribute to productivity outcomes has not been demonstrated.

Binary classification scope: The decision to implement binary rather than multi-class classification represents a deliberate simplification that limits diagnostic specificity. While this approach reduces complexity and supports the screening use case, it prevents the system from distinguishing between different diseases or providing disease-specific information that might inform treatment decisions.

Scalability uncertainty: Although the architecture was designed with scalability considerations, the system has not been tested under large-scale concurrent usage, high-volume image processing, or extended operational periods. Infrastructure requirements and performance characteristics at scale remain unknown.

Data quality dependencies: System performance is highly dependent on image quality, which in turn depends on factors such as camera resolution, lighting conditions, leaf positioning, and user capture technique. Poor quality images may significantly degrade classification accuracy, and the system currently lacks robust mechanisms for image quality assessment or user guidance.

Temporal and environmental variability: Coffee leaf appearance varies with plant age, nutritional status, environmental stress, and seasonal factors. The extent to which the trained models generalize across these sources of variation has not been systematically evaluated.

These limitations do not diminish the value of the work as a proof of concept and foundation for future development, but they do establish appropriate boundaries for claims regarding effectiveness and readiness for widespread deployment.

Market and Business Viability

From a market perspective, the exploratory validation process provided encouraging signals regarding problem relevance and solution fit. Conversations with individuals familiar with agricultural operations confirmed that monitoring scalability is a recognized challenge, particularly in large plantations with limited labor availability. The binary screening approach was perceived as more trustworthy and adoptable than fully automated diagnostic systems, supporting the project's conservative positioning.

The competitive analysis revealed that CropTrack occupies a distinct position between simple mobile diagnostic tools and comprehensive enterprise agritech platforms. Its focus on spatial integration combined with operational simplicity represents a differentiated value proposition that may resonate with medium-scale producers and agricultural service providers seeking incremental digitalization rather than transformative technological investment.

However, several business risks remain unresolved. The absence of commercial deployment means that critical metrics such as customer acquisition cost, lifetime value, and retention rates have not been empirically measured. User willingness to pay for binary screening functionality, as opposed to more comprehensive diagnostic capabilities, remains uncertain. Competition from established platforms with greater resources and market presence poses a persistent threat, particularly if those platforms introduce similar functionality as incremental features within broader product suites.

The financial structure of CropTrack, characterized by low marginal costs and subscription-based revenue logic, suggests long-term viability under conservative assumptions. However, achieving break-even depends heavily on adoption speed and retention, both of which require field validation and iterative product refinement based on real user feedback.

Future Developments and Research Directions

The foundation established by CropTrack creates multiple pathways for future development, both technical and operational.

Dataset expansion and diversification: The most immediate technical priority is the collection and integration of field-captured images representing real production conditions. Expanding the training dataset to include diverse environmental contexts, imaging devices, and disease progression stages would significantly improve model robustness and generalization. Establishing data collection partnerships with producers or research institutions could facilitate this expansion while validating the system under controlled field trials.

Multi-class disease classification: Extending the binary classification approach to distinguish between specific diseases would increase diagnostic value. This evolution could proceed incrementally, first adding a small number of high-priority disease classes and gradually expanding based on data availability and model performance. Such an extension would require careful attention to class imbalance, confusion between visually similar diseases, and the interpretability of multi-class predictions.

Integration with automated image capture: Rather than relying exclusively on manual image uploads, future versions could integrate with drones, automated imaging systems, or sensor networks deployed within fields. This integration would enable continuous or scheduled monitoring without requiring manual intervention, significantly increasing coverage and reducing operational friction.

Enhanced spatial analytics: The current aggregation of field-level health indicators could be extended to support temporal trend analysis, anomaly detection at sub-field

resolutions, and predictive modeling of disease spread based on historical patterns and environmental data. Integrating weather data, soil characteristics, and treatment history would enrich the spatial decision-support capabilities of the platform.

User experience refinement: Field validation will inevitably reveal usability issues and workflow misalignments that are not apparent during development. Iterative refinement based on producer feedback could address these issues and improve adoption rates. Features such as guided image capture, real-time quality feedback, and simplified result interpretation may prove critical for practical deployment.

Explainability and interpretability: Incorporating visual explanation techniques such as gradient-based attention maps or saliency visualization could help users understand why the model classified a particular image as unhealthy. This transparency may increase trust and enable producers to learn visual disease indicators, complementing the automated screening functionality.

Integration with existing farm management systems: Rather than operating as a standalone platform, CropTrack could be integrated with established farm management software through APIs or modular plugins. This integration strategy would reduce adoption barriers and leverage existing user bases, albeit at the cost of reduced control over the complete user experience.

Economic impact assessment: Conducting formal studies to quantify the economic value generated by early anomaly detection would strengthen the business case for adoption. Measuring metrics such as reduction in disease-related losses, efficiency gains in inspection workflows, and optimization of treatment interventions would provide empirical support for the system's value proposition.

Regulatory and certification considerations: As the system matures toward commercial deployment, addressing data privacy regulations, agricultural certification requirements, and potential liability concerns will become increasingly important. Proactive engagement with these considerations will reduce future compliance risk.

Final Considerations

CropTrack represents a deliberate effort to bridge the gap between academic machine learning research and practical agricultural application. The project demonstrates that computer vision techniques can be effectively integrated into decision-support workflows that respect the spatial nature of agricultural reasoning and the importance of human expertise in agronomic judgment.

The choice to focus on binary health screening rather than comprehensive diagnosis reflects a pragmatic understanding of the constraints inherent in early-stage development: limited data, academic time horizons, and the need to establish technical feasibility before pursuing functional completeness. This approach prioritizes learning through implementation and positions the project as a foundation for iterative evolution rather than a finished product.

The modular architecture and transparent methodology enable future researchers and developers to extend, adapt, or critique the work without encountering unnecessary barriers. By documenting design decisions, acknowledging limitations, and grounding conclusions in empirical validation, the project contributes to the responsible advancement of AI applications in agriculture.

Ultimately, CropTrack's value lies not in claiming to solve the full complexity of agricultural disease management, but in demonstrating that incremental, well-scoped applications of artificial intelligence can provide meaningful support to producers facing real operational challenges. The system augments rather than replaces human expertise, increases monitoring coverage without eliminating the need for field knowledge, and provides spatial context that connects automated analysis to actionable decisions.

The path from minimum viable product to widely adopted agricultural tool remains long and uncertain. However, the technical foundation, market validation insights, and clearly articulated limitations established by this work provide a credible starting point for that journey. Whether CropTrack evolves into a commercial product, informs other agricultural AI initiatives, or serves primarily as an educational reference for applied engineering projects, its contribution lies in demonstrating that thoughtful

integration of machine learning into agricultural workflows is both technically feasible and operationally valuable.

The future of precision agriculture will likely involve increasingly sophisticated combinations of sensing technologies, data analytics, and automated decision-support. CropTrack's focused, transparent, and user-centered approach offers one model for how such systems might be developed responsibly and deployed effectively, with appropriate respect for the complexity of agricultural practice and the irreplaceable value of agronomic expertise.

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